```
data <- iris
percent <- as.numeric(
  readline("Enter training percentage: ")
)
set.seed(10)
n <- nrow(data)
train_n <- round(n * percent / 100)
index <- sample(1:n, train_n)
train <- data[index, ]
test <- data[-index, ]
# Logistic model for Setosa
m1 <- glm(
  I(Species == "setosa") ~ Petal.Length + Petal.Width,
  data = train,
  family = binomial
)
# Logistic model for Versicolor
m2 <- glm(
  I(Species == "versicolor") ~ Petal.Length + Petal.Width,
  data = train,
  family = binomial
)
# Logistic model for Virginica
m3 <- glm(
  I(Species == "virginica") ~ Petal.Length + Petal.Width,
  data = train,
```

```r
  family = binomial
)
# Get probabilities
p1 <- predict(m1, test, type = "response")
p2 <- predict(m2, test, type = "response")
p3 <- predict(m3, test, type = "response")
# Select highest probability
result <- cbind(p1, p2, p3)
prediction <- apply(
  result,
  1,
  which.max
)

prediction <- c(
  "setosa",
  "versicolor",
  "virginica"
)[prediction]
# Confusion matrix
cm <- table(
  Actual = test$Species,
  Predicted = prediction
)
cat("\nConfusion Matrix:\n")
print(cm)
```

```
# Accuracy

accuracy <- mean(

  prediction == test$Species

)

cat("\nAccuracy:",

    round(accuracy * 100, 2),

    "%")
```